Linear inequality in one variable

A linear inequality which has only one variable is called linear inequality in one variable.

e.g. $ax + b < 0$, where $a \neq \left(\frac{-b}{a}\right)$, $4x + 7 \geq 0$

Linear inequality in two variables

A linear inequality which has two variables, is called linear inequality in two variables.

e.g. $3x + 11y \leq 0$, $4t + 3y > 0$

General Rules to Solve Linear Inequalities

- Equal numbers may be added or subtracted from both sides of an inequality without changing the sign of in equality.
  e.g. If $a > b$, then for any number c,
  $a + c > b + c$ or $a - c > b - c$.
- If both sides of an inequality are multiplied or divided by the same positive number, then the sign of inequality remains the same. e.g.
  If $a > b$ and $c > 0$, then $\frac{a}{c} > \frac{b}{c}$ and $ac > bc$.
- If both sides of an inequality are multiplied or divided by the same negative number, then the sign of inequality is reversed.
  e.g. If $a > b$ and $c < 0$, then $\frac{a}{c} < \frac{b}{c}$ and $ac < bc$.

Graphical Solution of Linear Inequalities in Two Variables

Let the inequality be $ax + by + c \geq 0$.

Step I Consider it as $ax + by + c = 0$ by changing the inequality sign to equality sign, Draw the graph of this equation.

Step II Choose any point not on the line, [usually $(0, 0)$].

Find whether this point satisfy the inequality or not. If it does, then shade the half-plane containing that point. Otherwise shade the other half-plane.

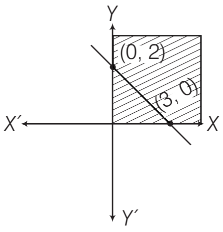
EXAMPLE 20. The solution set of inequality $2x + 3y \geq 6$, $x \geq 0$ and $y \geq 0$ is

a. $x = 1, y = 1$ b. $x = -1, y = -1$
c. $x = 3, y = 1$ d. None of these

Sol. d. Draw the graph of $2x + 3y = 6$.

The values of (x, y) satisfying $2x + 3y = 6$ are

x	0	3
y	2	0



Clearly, here (0, 0) does not satisfies the inequality, so we shade the plane which does not contain (0, 0).

Also, since $x \geq 0$ and $y \geq 0$, so the solution set must be in Ist quadrant only.

Hence, the shaded region represents the solution set of the given system of inequalities.

Note (i) If the inequality is strict (> or <), graph a dashed line. If the inequality is not strict (≥ or ≤), graph a solid line.

(ii) In order to solve a system of two or more linear inequalities with the same variables, graph each inequality on the same set of axes. The intersection of the region of all the inequalities is the required solution set.

EXAMPLE 21. Find the solution set of $-2x + 3y > 6$ and $4x - 6y > 12$.

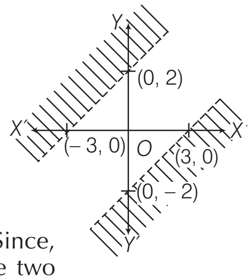
a. $x < -1, y > -3$ b. $x < 1, y > 3$
c. $x \geq 3, y \geq 2$ d. No solution

Sol. d. Consider, $-2x + 3y = 6$

x	0	-3
y	2	0

Again, $4x - 6y = 12$

x	0	3
y	-2	0



Now, graph each equation. Since, there is no intersection of these two shaded regions, therefore there is no solution.

Quadratic Inequalities

An equation of the form

$$ax^2 + bx + c \geq 0 \quad \text{or} \quad ax^2 + bx + c \leq 0$$

or

$$ax^2 + bx + c > 0 \quad \text{or} \quad ax^2 + bx + c < 0$$

where, $a \neq 0$ is called a quadratic inequality in one variable x. It is same as quadratic equations except for the inequality sign.

Solution of Quadratic Inequalities

Solving the inequality means finding the values of x that make the inequality true. Solution sets of quadratic inequalities are expressed in the form of intervals. Here are the steps:

- Replace the inequality symbol with an equal sign.
- Solve the quadratic equation by factorisation or by quadratic formula and find the real roots of the equation.
- Plot the two real roots on the numbers line. They divide the line into three sections or three intervals.

- Pick a number from each interval and test it in the original inequality.
- If the inequality holds true for the choosen point. then that interval is the solution of the given quadratic inequality.

Note (i) If the quadratic equation does not have real roots, then the quadratic expression is always positive (or always negative) depending on the sign of a meaning that the solution set will either be empty or the entire real number line.

e.g. $15x^2 - 18x + 7 > 0$ has no real roots as $D = b^2 - 4ac = 64 - 420 < 0$. Since $a > 0$, so the expression is always positive and hence the solution set will be the entire real number line.

(ii) When the inequality has an additional '=' sign (≥ or ≤), use closed intervals like $[a, b]$ to indicate that the two end points are also included in the solution set. If the inequality is strict (>or<), use open intervals like (a, b) as end points are not included.

EXAMPLE 22. The solution of the inequality $x^2 + 4x + 3 \geq 0$ is

a. $x < -3$ and $x > -1$ b. $x \leq -3$ and $x \geq -1$

c. $x < -3$ and $x \geq -1$ d. $x \leq -3$ or $x > -1$

Sol. b. We have, $x^2 + 4x + 3 \geq 0$

Change the inequality to equality sign and solve

$$x^2 + 4x + 3 = 0 \Rightarrow x^2 + 3x + x + 3 = 0$$

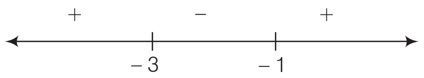
$$\Rightarrow x(x+3) + (x+3) = 0 \Rightarrow (x+1)(x+3) = 0 \Rightarrow x = -1, -3.$$

Place the value of x on the number line to create intervals.

Take $x = -4$, Take $x = -2$, Take $x = 0$

$\Rightarrow (-4)^2 + 4(-4) + 3 > 0$,

$(-2)^2 + 4(-2) + 3 < 0$, $0^2 + 4(0) + 3 > 0$



So, the intervals which satisfy the given equation are $(-\infty, -3]$ and $[-1, \infty)$

i.e. $x \leq -3$ and $x \geq -1$

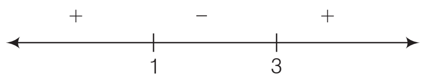
EXAMPLE 23. The real values of x which satisfy $x^2 - 4x + 3 \geq 0$ and $x^2 - 3x - 4 \leq 0$ is

a. $(-1, 1) \cup (3, 4)$ b. $[-1, 1] \cap [3, 4]$

c. $[-1, 1] \cup [3, 4]$ d. $(-1, 1) \cup [3, 4]$

Sol. c. Given, $x^2 - 4x + 3 \geq 0$

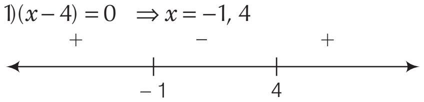
$\Rightarrow \quad x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0 \Rightarrow x = 1, 3$



$\Rightarrow x \in (-\infty, 1]$ or $x \in [3, \infty)$. $\Rightarrow x \in (-\infty\ 1] \cup [3, \infty)$.

Again, $x^2 - 3x - 4 \leq 0 \Rightarrow x^2 - 3x - 4 = 0$

$\Rightarrow \quad (x+1)(x-4) = 0 \Rightarrow x = -1, 4$



$\Rightarrow x \in [-1, 4]$

$\therefore$ Solution set is

$x = (-\infty, 1] \cup [3, \infty) \cap [-1, 4] = [-1, 1] \cup [3, 4]$

PRACTICE EXERCISE

1. The quadratic equation has maximum

(a) one root (b) two roots

(c) four roots (d) three roots

2. The values of x in the equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0$, $a \neq 0$, $b \neq 0$ is

(a) $1/a^2$ (b) $1/b^2$

(c) $1/a^2, 1/b^2$ (d) None of these

3. The value of 'a' for which the equation $ax^2 - 2\sqrt{5}\,x + 4 = 0$ has equal roots is

(a) 5/4 (b) 4/5

(c) − 5/4 (d) − 5/3

4. If one root of $3x^2 = 8x + (2k+1)$ is seven times the other, then the value of k is

(a) 5/3 (b) − 5/3

(c) 2/3 (d) − 3/2

5. The quadratic equation whose roots are $\frac{4+\sqrt{7}}{2}$ and $\frac{4-\sqrt{7}}{2}$ is

(a) $4x^2 + 16x + 9 = 0$ (b) $4x^2 - 16x - 9 = 0$

(c) $4x^2 - 16x + 9 = 0$ (d) $4x^2 + 16x - 9 = 0$

6. If α and β are the roots of the equation $x^2 - 8x + p = 0$ and $\alpha^2 + \beta^2 = 40$, then p is equal to

(a) 12 (b) 10 (c) 9 (d) 11

7. If α and β are the roots of the equation $x^2 - 5x + 6 = 0$, then the value of $\alpha^2 - \beta^2$

(a) 5 (b) − 5 (c) ± 5 (d) ± 4

8. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then an equation whose roots are $1/\alpha$ and $1/\beta$ is

(a) $bx^2 + ax + c = 0$ (b) $ax^2 - bx + c = 0$

(c) $cx^2 + ax + b = 0$ (d) $cx^2 + bx + a = 0$

9. If α, β are the roots of a quadratic equation such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$, then the equation is

(a) $x^2 - 24x - 128 = 0$ (b) $x^2 + 24x + 128 = 0$
(c) $x^2 + 24x - 128 = 0$ (d) None of these

10. Which one of the following is the equation whose roots are respectively three times the roots of the equation $ax^2 + bx + c = 0$?

(a) $ax^2 + bx + c = 0$ (b) $ax^2 + 3bx + 9c = 0$
(c) $ax^2 - 3bx + 9c = 0$ (d) $ax^2 + bx + 3c = 0$

11. How many real values of x satisfy the equation $x^{2/3} + x^{1/3} - 2 = 0$?

(a) only one value (b) two values
(c) three values (d) No value

12. If α, β are the roots of the quadratic equation $2x^2 - 4x + 1 = 0$. Then, the value of $\frac{1}{\alpha + 2\beta} + \frac{1}{\beta + 2\alpha}$ is equal to

(a) $\frac{12}{17}$ (b) $\frac{17}{12}$ (c) $\frac{11}{17}$ (d) $\frac{13}{17}$

13. Solve the equation $2^{x+2} + 2^{-x} = 5$ and find the roots of the equation.

(a) $\{0, 2\}$ (b) $\{-2, 0\}$ (c) $\{0, 1\}$ (d) $\{-1, 0\}$

14. If α and β are roots of the equation $x^2 + p = 0$ where p is a prime, then which equation has the roots $1/\alpha$ and $1/\beta$?

(a) $\frac{1}{x^2} - \frac{1}{p} = 0$ (b) $px^2 + 1 = 0$ (c) $px^2 - 1 = 0$ (d) $\frac{1}{x^2} + \frac{1}{p} = 0$

15. The roots of the equation $x^2 + px + q = 0$ are 1 and 2. The roots of the equation $qx^2 - px + 1 = 0$ must be

(a) $\frac{-1}{2}$ and 1 (b) $\frac{1}{2}$ and 1
(c) $\frac{-1}{2}$ and -1 (d) None of these

16. Which one of the following is the quadratic equation whose roots are reciprocal to the roots of the quadratic equation $2x^2 - 3x - 4 = 0$?

(a) $3x^2 - 2x - 4 = 0$ (b) $4x^2 + 3x - 2 = 0$
(c) $3x^2 - 4x - 2 = 0$ (d) $4x^2 - 2x - 3 = 0$

17. The value of x satisfying the equation $\sqrt{x+4} = x - 2$ is

(a) 0, 5 (b) 0, 4
(c) 5 (d) None of these

18. If the roots of the quadratic equation $px^2 + qx + r = 0$ are reciprocal to each other, then

(a) $q = r$ (b) $p = r$
(c) q divides r (d) p divides q

19. Sum of roots is -1 and sum of their reciprocals is 1/6, then equation is

(a) $x^2 - 6x + 1 = 0$ (b) $x^2 - x + 6 = 0$
(c) $6x^2 + x + 1 = 0$ (d) $x^2 + x - 6 = 0$

20. If sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of their squares, then which one of the following is correct?

(a) $a^2 + b^2 = c^2$ (b) $a^2 + b^2 = a + b$
(c) $2ac = ab + b^2$ (d) $2c + b = 0$

21. If the roots of $x^2 - lx + m = 0$ differ by 1, then

(a) $l^2 = 4m - 1$ (b) $l^2 = 4m + 2$
(c) $l = 4m^2 + 1$ (d) $l^2 = 4m + 1$

22. If α, β are the roots of the equation $x^2 - (1 + a^2)\,x + \frac{1}{2}(1 + a^2 + a^4) = 0$, then $\alpha^2 + \beta^2$ is equal to

(a) $a^4 + a^2$ (b) a^2
(c) $(a^2 + a^4)^2$ (d) None of these

23. If the roots of $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude and opposite in sign, then product of roots is

(a) $-\frac{1}{2}(p^2 + q^2)$ (b) $\frac{p^2 + q^2}{2}$
(c) $\frac{p+q}{2}$ (d) $\frac{1}{2}(p+q)^2$

24. If α, β are the roots of $3x^2 + 2x + 1 = 0$, then the equation whose roots are $\frac{1-\alpha}{1+\alpha}$ and $\frac{1-\beta}{1+\beta}$ is

(a) $x^2 + 2x + 3 = 0$ (b) $x^2 - 2x + 3 = 0$
(c) $x^2 + 2x - 3 = 0$ (d) $x^2 - 2x - 3 = 0$

25. If one root of $px^2 + qx + r = 0$ is double of the other root, then which one of the following is correct?

(a) $2q^2 = 9pr$ (b) $2q^2 = 9p$ (c) $4q^2 = 9r$ (d) $9q^2 = 2pr$

26. If the roots of the equation $x^2 + x + 1 = 0$ are in the ratio $m : n$, then

(a) $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = 0$ (b) $\sqrt{m} + \sqrt{n} + 1 = 0$
(c) $\frac{m}{n} + \frac{n}{m} + 1 = 0$ (d) $m + n + 1 = 0$

27. What is one of the value of x in the equation $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$?

(a) $\frac{5}{13}$ (b) $\frac{7}{13}$ (c) $\frac{9}{13}$ (d) $\frac{11}{3}$

28. What are the roots of the equation
$$(a+b+x)^{-1} = a^{-1} + b^{-1} + x^{-1}\ ?$$
(a) a, b (b) $-a, b$ (c) $a, -b$ (d) $-a, -b$

29. The number of roots of the quadratic equation $8\sec^2\phi - 6\sec\phi + 1 = 0$ is
(a) n (b) 2 (c) 0 (d) No solution

30. The number of real roots of $3^{2x^2-7x+7} = 9$ is
(a) 3 (b) 1 (c) 2 (d) 4

31. If $x = \sqrt{2} + 2$, then
(a) $x^2 + 4x + 2 = 0$ (b) $x^2 - 2x - 2 = 0$
(c) $x^2 - 4x + 2 = 0$ (d) $x^2 - 4x - 2 = 0$

32. For what value of k will the equation $\frac{x^2 - bx}{ax - c} = \frac{k-1}{k+1}$ have roots reciprocal to each other?
(a) $\frac{1+c}{1-c}$ (b) $\frac{c+1}{c-1}$ (c) $a - c$ (d) $\frac{b+c}{c-1}$

33. If 'α' and 'β' be the roots of $ax^2 - bx + b = 0$, the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$ is
(a) a/b (b) $\sqrt{b/a}$ (c) $\sqrt{a/b}$ (d) $-a/b$

34. If the equations $2x^2 - 7x + 3 = 0$ and $4x^2 + ax - 3 = 0$ have a common root, then what is the value of a?
(a) -11 or 4 (b) -11 or -4 (c) 11 or -4 (d) 11 or 4

35. If the roots of $x^2 + bx + c = 0$ be α and β and those of $x^2 + px + q = 0$ be $k\alpha$ and $k\beta$, then
(a) $cb^2 = qp^2$ (b) $qc^2 = b^2p^2$
(c) $qb^2 = cp^2$ (d) None of these

36. If -4 is a root of the equation $x^2 + px - 4 = 0$ and the equation $x^2 + px + q = 0$ has equal roots, then the values of p and q are, respectively
(a) -3 and $\frac{9}{4}$ (b) 3 and $\frac{9}{4}$ (c) $\frac{9}{4}$ and 3 (d) 4 and 3

37. If α, β are the roots of $2x^2 - 6x + 3 = 0$, then the value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 3\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 2\alpha\beta$ is
(a) 12 (b) 23 (c) 13 (d) -13

38. If the roots of the equation $x^2 + 2ax + b = 0$, are real and distinct and they differ by atmost 2m, then b lies in the interval
(a) $(a^2 - m^2, a^2)$ (b) $[a^2 - m^2, a^2)$
(c) $(a^2, a^2 + m^2)$ (d) None of these

39. The solution of the equation
$$(x+2)(x-5)(x-6)(x+1) = 144 \text{ is}$$
(a) $\{7, -3\}$ (b) $\{7, -3, 2\}$ (c) $\{7, -3, 2, 1\}$ (d) $\{7, 2\}$

40. The solution of the equation $\sqrt{x^2 - 16} - (x-4) = \sqrt{x^2 - 5x + 4}$ is
(a) $\left\{4, 5, -\frac{13}{3}\right\}$ (b) $\{4, 5\}$ (c) $\{4\}$ (d) $\left\{5, -\frac{13}{3}\right\}$

41. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then which one of the following is correct?
(a) $a^2 + b^2 + 2ac = 0$ (b) $a^2 - b^2 + 2ac = 0$
(c) $a^2 + c^2 + 2ab = 0$ (d) $a^2 - b^2 - 2ac = 0$

42. The equation $(1+n^2)x^2 + 2ncx + (c^2 - a^2) = 0$ will have equal roots, if
(a) $c^2 = 1 + a^2$ (b) $c^2 = 1 - a^2$
(c) $c^2 = 1 + n^2 + a^2$ (d) $c^2 = (1+n^2)a^2$

43. What is the condition that the equation $ax^2 + bx + c = 0$, where $a \neq 0$ has both the roots positive?
(a) a, b and c are of same sign
(b) a and b are of same sign
(c) b and c have the same sign opposite to that of a
(d) a and c have the same sign opposite to that of b

44. The equation whose roots are twice the roots of the equation $x^2 - 2x + 4 = 0$ is
(a) $x^2 - 2x + 4 = 0$ (b) $x^2 - 2x + 16 = 0$
(c) $x^2 - 4x + 8 = 0$ (d) $x^2 - 4x + 16 = 0$

45. If α and β are the roots of the equation $x^2 + px + q = 0$, then $-\alpha^{-1}, -\beta^{-1}$ are the roots of which one of the following equations?
(a) $qx^2 - px + 1 = 0$ (b) $q^2 + px + 1 = 0$
(c) $x^2 + px - q = 0$ (d) $x^2 - px + q = 0$

46. If one root of the equation $ax^2 + x - 3 = 0$ is -1, then what is the other root?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

47. If the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ has equal roots, then which one of the following is correct?
(a) $ab = cd$ (b) $ad = bc$
(c) $a^2 + c^2 = b^2 + d^2$ (d) $ac = bd$

48. What is the solution of the equation
$$\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}\,?$$
(a) 1 (b) 2 (c) 4 (d) None of these

49. What are the roots of the equation
$4^x - 3 \cdot 2^{x+2} + 32 = 0$?
(a) 1, 2 (b) 3, 4 (c) 2, 3 (d) 1, 3

50. If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is the value of $(\alpha^4 + \beta^4)$?
(a) 7 (b) 0
(c) 2 (d) None of these

51. When the roots of the quadratic equation $ax^2 + bx + c = 0$ are negative and reciprocals of each other, then which one of the following is correct?
(a) $b = 0$ (b) $c = 0$ (c) $a = c$ (d) $a = -c$

52. If sum as well as product of roots of a quadratic equation is 9, then what is the equation?
(a) $x^2 + 9x - 18 = 0$ (b) $x^2 - 18x + 9 = 0$
(c) $x^2 + 9x + 9 = 0$ (d) $x^2 - 9x + 9 = 0$

53. What are the roots of the equation $\log_{10}(x^2 - 6x + 45) = 2$?
(a) 9, – 5 (b) – 9, 5 (c) 11, – 5 (d) – 11, 5

54. The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. What is the product of the roots of the equation?
(a) $-\frac{(a+b)}{2}$ (b) $\frac{(a+b)}{2}$
(c) $-\frac{(a^2+b^2)}{2}$ (d) $\frac{(a^2+b^2)}{2}$

55. For what value of k, will the roots of the equation $kx^2 - 5x + 6 = 0$ be in the ratio of 2 : 3?
(a) 0 (b) 1 (c) – 1 (d) 2

56. What is the ratio of sum of squares of roots to the product of the roots of the equation $7x^2 + 12x + 18 = 0$?
(a) 6 : 1 (b) 1 : 6 (c) – 6 : 1 (d) – 6 : 7

57. If the roots of the equation $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$ are equal, then what is the value of m?
(a) 1 (b) 1/2 (c) 0 (d) – 1/2

58. If $3^x + 27(3^{-x}) = 12$, then what is the value of x?
(a) 1 (b) 2 (c) 1, 2 (d) 0, 1

59. What is the magnitude of difference of the roots of $x^2 - ax + b = 0$?
(a) $\sqrt{a^2 - 4b}$ (b) $\sqrt{b^2 - 4a}$
(c) $2\sqrt{a^2 - 4b}$ (d) $\sqrt{b^2 - 4ab}$

60. What can be said about the roots of the equation $x^2 - x - 2 = 0$?
(a) both of them are integers
(b) both of them are natural numbers
(c) the latter of the two is negative
(d) None of the above

61. An equation equivalent to the quadratic equation $x^2 - 6x + 5 = 0$ is
(a) $x^2 - 5x + 6 = 0$ (b) $5x^2 - 6x + 1 = 0$
(c) $|x - 3| = 2$ (d) $6x^2 - 5x + 1 = 0$

62. If the sum of a number and its reciprocal is $\frac{10}{3}$, then the numbers are
(a) $3, \frac{1}{3}$ (b) $3, -\frac{1}{3}$ (c) $-3, \frac{1}{3}$ (d) $-3, -\frac{1}{3}$

63. Divide 16 into two parts such that the twice of the square of the greater part exceeds, the square of the smaller part by 164. Then, the greater part is
(a) 58 (b) 10 (c) 6 (d) 15

64. The number of straight lines that can connect 'x' points is given by the equation $y = \frac{x(x-1)}{2}$. How many points does a figure have if only 15 lines can be drawn connecting them?
(a) 15 (b) 10 (c) 6 (d) 5

65. The two successive natural numbers whose squares have sum 221 are
(a) 10 and 11 (b) 11 and 12
(c) – 10 and – 11 (d) None of these

66. The two consecutive positive odd integers, the sum of whose squares is 130.
(a) – 7 and – 9 (b) 7 and 9
(c) 7 and 5 (d) 3 and – 5

67. The solution set of inequation $2x + 1 \geq 7$ is
(a) $x \geq 8$ (b) $x \geq 3$
(c) $x \geq 6$ (d) None of these

68. The values of x satisfying inequation $\frac{x-1}{3} \geq 4$ is
(a) $x \leq 13$ (b) $x \geq 12$ (c) $x \geq 13$ (d) $x = 13$

69. The values of x satisfying $3x + 2 \leq 5x - (4 - x)$ is
(a) $x \leq 2$ (b) $x \geq 2$
(c) $x = 2$ (d) None of these

70. The solution set of x for the inequations $2x + 3 \geq 8$ and $3x + 1 \leq 12$ is
(a) $\frac{5}{2} < x \leq \frac{11}{3}$ (b) $\frac{5}{2} < x < \frac{11}{3}$
(c) $\frac{5}{2} \leq x \leq \frac{11}{3}$ (d) $\frac{5}{2} \geq x \geq \frac{11}{3}$

71. The values of x satisfying $\frac{1}{2}\left(\frac{3}{5}x + 4\right) \geq \frac{1}{3}(x - 6)$ are
(a) $x \geq 120$ (b) $x \leq 120$ (c) $x \leq 12$ (d) $x \geq 12$

72. $4x^2 - 1 \leq 0$, then the solution set is
(a) $\frac{-1}{2} \leq x \leq \frac{1}{2}$ (b) $\frac{-1}{2} < x < \frac{1}{2}$
(c) $\frac{-1}{2} \geq x \geq \frac{1}{2}$ (d) None of these